Review

80%

37
Out of 46 points

2:26
Avg Time Per Question

Your Answers:

Vignette

Viktoria Smith is a recently hired junior analyst at Aries Investments. Smith and her supervisor, Ingrid Johansson, meet to discuss some of the firm's investments in banks and insurance companies.

Johansson asks Smith to explain why the evaluation of banks is different from the evaluation of non-financial companies. Smith tells Johansson the following:

Statement 1:

As intermediaries, banks are more likely to be systemically important than non-financial companies.

Statement 2:

The assets of banks mostly consist of deposits, which are exposed to different risks than the tangible assets of non-financial companies.

Smith and Johansson also discuss key aspects of financial regulations, particularly the framework of Basel III. Johansson tells Smith:

> *"Basel III specifies the minimum percentage of its risk-weighted assets that a bank must fund with equity. This requirement of Basel III prevents a bank from assuming so much financial leverage that it is unable to withstand loan losses or asset write-downs."*

Johansson tells Smith that she uses the CAMELS approach to evaluate banks, even though it has some limitations. To evaluate P&C insurance companies, Johansson tells Smith that she places emphasis on the efficiency of spending on obtaining new premiums. Johansson and Smith discuss differences between P&C and L&H insurance companies. Smith notes the following differences:

Difference 1:

L&H insurers' claims are more predictable than P&C insurers' claims.

Difference 2:

P&C insurers' policies are usually short term, whereas L&H insurers' policies are usually longer term.

Difference 3:

Relative to L&H insurers, P&C insurers often have lower capital requirements and can also seek higher returns offered by riskier investments.

Johansson asks Smith to review key performance ratios for three P&C insurers in which Aries is invested. The ratios are presented in Exhibit 1.

Exhibit 1
Key Performance Ratios for Selected P&C Insurers

	Insurer A	**Insurer B**	**Insurer C**
Loss and loss adjustment expense ratio	68.8%	65.9%	64.1%
Underwriting expense ratio	33.7%	37.8%	32.9%
Combined ratio	102.5%	103.7%	97.0%

Johansson also asks Smith to review key performance ratios for ABC Bank, a bank in which Aries is invested. The ratios are presented in Exhibit 2.

Exhibit 2
Key Performance Ratios for ABC Bank*

	2017	**2016**	**2015**
Common equity Tier 1 capital ratio	10.7%	11.5%	12.1%
Tier 1 capital ratio	11.5%	12.6%	13.4%
Total capital ratio	14.9%	14.8%	14.9%
Liquidity coverage ratio	123.6%	121.4%	119.1%

	2017	2016	2015
Net stable funding ratio	114.9%	113.2%	112.7%
Total trading VaR (all market risk factors)	$11	$13	$15
Total trading and credit portfolio VaR	$15	$18	$21

* *Note*: VaR amounts are in millions and are based on a 99% confidence interval and a single-day holding period.

1 Multiple Choice 1 / 1 point

Which of Smith's statements regarding banks is correct?

- A. Only Statement 1
- B. Only Statement 2
- C. Both Statement 1 and Statement 2

Feedback

A is correct. Banks are more likely to be systemically important than non-financial companies because, as intermediaries, they create financial linkages across all types of entities, including households, banks, corporates, and governments. The network of linkages across entities means that the failure of one bank will negatively affect other financial and non-financial entities (a phenomenon known as financial contagion). The larger the bank and the more widespread its network of linkages, the greater its potential impact on the entire financial system. The assets of banks are predominantly financial assets, such as loans and securities (not deposits, which represent most of a bank's liabilities). Compared to the tangible assets of non-financial companies, financial assets create direct exposure to a different set of risks, including credit risks, liquidity risks, market risks, and interest rate risks.

2 Multiple Choice 1 / 1 point

The aspect of the Basel III framework that Johansson describes to Smith relates to minimum:

- A. capital requirements.
- B. liquidity requirements.
- C. amounts of stable funding requirements.

Feedback

A is correct. Basel III specifies the minimum percentage of its risk-weighted assets that a bank must fund with equity capital. This minimum funding requirement prevents a bank from assuming so much financial leverage that it is unable to withstand loan losses or asset write-downs.

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3 Multiple Choice 1 / 1 point

One limitation of the approach used by Johansson to evaluate banks is that it fails to address a bank's:

A. sensitivity to market risk.

B. management capabilities.

C. competitive environment.

Feedback

C is correct. The approach used by Johansson to evaluate banks, the CAMELS approach, has six components: (1) capital adequacy, (2) asset quality, (3) management capabilities, (4) earnings sufficiency, (5) liquidity position, and (6) sensitivity to market risk. While the CAMELS approach to evaluating a bank is fairly comprehensive, some attributes of a bank are not addressed by this method. One such attribute is a bank's competitive environment. A bank's competitive position relative to its peers may affect how it allocates capital and assesses risks.

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4 Multiple Choice 1 / 1 point

The best indicator of the operations of a P&C insurance company emphasized by Johansson when evaluating P&C insurance companies is the:

A. combined ratio.

B. underwriting loss ratio.

C. underwriting expense ratio.

Feedback

C is correct. The underwriting expense ratio is an indicator of the efficiency of money spent on obtaining new premiums. The underwriting loss ratio is an indicator of the quality of a company's underwriting activities—the degree of success an underwriter has achieved in estimating the risks insured. The combined ratio, a measure of the overall underwriting profitability and efficiency of an underwriting operation, is the sum of these two ratios.

5 Multiple Choice 1 / 1 point

Which of the differences between P&C insurers and L&H insurers noted by Smith is *incorrect*?

A. Difference 1

B. Difference 2

C. Difference 3

Feedback

C is correct. The products of the two types of insurance companies, P&C and L&H, differ in contract duration and claim variability. P&C insurers' policies are usually short term, and the final cost will usually be known within a year of the occurrence of an insured event, while L&H insurers' policies are usually longer term. P&C insurers' claims are more variable and "lumpier" because they arise from accidents and other less predictable events, while L&H insurers' claims are more predictable because they correlate closely with relatively stable, actuarially based mortality rates applied to large populations. The relative predictability of L&H insurers' claims generally allows these companies to have lower capital requirements and to seek higher returns than P&C insurers.

6 Multiple Choice 1 / 1 point

Based on Exhibit 1, Smith should conclude that the insurer with the most efficient underwriting operation is:

A. Insurer A.

B. Insurer B.

C. Insurer C.

Feedback

C is correct. The combined ratio, which is the sum of the underwriting expense ratio and the loss and loss adjustment expense ratio, is a measure of the efficiency of an underwriting operation. A combined ratio of less than 100% is considered efficient; a combined ratio greater than 100% indicates an underwriting loss. Insurer C is the only insurer that has a combined ratio less than 100%.

7 Multiple Choice 1 / 1 point

Based on Exhibit 2, Smith and Johansson should conclude that over the past three years, ABC Bank's:

A. liquidity position has declined.

B. capital adequacy has improved.

C. sensitivity to market risk has improved.

Feedback

C is correct. Over the past three years, there has been a downward trend in the two VaR measures—total trading VaR (all market risk factors) and total trading and credit portfolio VaR. This trend indicates an improvement in ABC Bank's sensitivity, or a reduction in its exposure, to market risk. The two liquidity measures—the liquidity coverage ratio and the net stable funding ratio—have increased over the past three years, indicating an improvement in ABC Bank's liquidity position. Trends in the three capital adequacy measures—common equity Tier 1 capital ratio, Tier 1 capital ratio, and total capital ratio—indicate a decline in ABC Bank's capital adequacy. While the total capital ratio has remained fairly constant over the past three years, the common equity Tier 1 capital ratio and the Tier 1 capital ratio have declined. This trend suggests that ABC Bank has moved toward using more Tier 2 capital and less Tier 1 capital, indicating an overall decline in capital adequacy.

Vignette

Judith Yoo is a financial sector analyst writing an industry report. In the report, Yoo discusses the relative global systemic risk across industries, referencing Industry A (international property and casualty insurance), Industry B (credit unions), and Industry C (global commercial banks).

Part of Yoo's analysis focuses on Company XYZ, a global commercial bank, and its CAMELS rating, risk management practices, and performance. First, Yoo considers the firm's capital adequacy as measured by the key capital ratios (common equity Tier 1 capital, total Tier 1 capital, and total capital) in Exhibit 1.

Exhibit 1

Company XYZ: Excerpt from Annual Report Disclosure

At 31 December	2017	2016	2015
Regulatory capital	**$m**	**$m**	**$m**
Common equity Tier 1 capital	146,424	142,367	137,100
Additional Tier 1 capital	22,639	20,443	17,600
Tier 2 capital	22,456	27,564	38,200
Total regulatory capital	191,519	190,374	192,900
Risk-weighted assets (RWAs) by risk type			
Credit risk	960,763	989,639	968,600
Market risk	44,100	36,910	49,600
Operational risk	293,825	256,300	224,300
Total RWAs	1,298,688	1,282,849	1,242,500

Yoo turns her attention to Company XYZ’s asset quality using the information in Exhibit 2.

Exhibit 2
Company XYZ: Asset Composition

At 31 December	2017	2016	2015
	$m	**$m**	**$m**
Total liquid assets	361,164	354,056	356,255
Investments	434,256	367,158	332,461
Consumer loans	456,957	450,576	447,493
Commercial loans	499,647	452,983	403,058

At 31 December	2017	2016	2015
	$m	$m	$m
Goodwill	26,693	26,529	25,705
Other assets	151,737	144,210	121,780
Total assets	1,930,454	1,795,512	1,686,752

To assess Company XYZ's risk management practices, Yoo reviews the consumer loan credit quality profile in Exhibit 3 and the loan loss analysis in Exhibit 4.

Exhibit 3
Company XYZ: Consumer Loan Profile by Credit Quality

At 31 December	2017	2016	2015
	$m	$m	$m
Strong credit quality	338,948	327,345	320,340
Good credit quality	52,649	54,515	54,050
Satisfactory credit quality	51,124	55,311	56,409
Substandard credit quality	23,696	24,893	27,525
Past due but not impaired	2,823	2,314	2,058
Impaired	8,804	9,345	10,235
Total gross amount	478,044	473,723	470,617
Impairment allowances	−5,500	−4,500	−4,000
Total	472,544	469,223	466,617

Exhibit 4
Company XYZ: Loan Loss Analysis Data

At 31 December	**2017**	**2016**	**2015**
	$m	**$m**	**$m**
Consumer loans			
Allowance for loan losses	11,000	11,500	13,000
Provision for loan losses	3,000	2,000	1,300
Charge-offs	3,759	3,643	4,007
Recoveries	1,299	1,138	1,106
Net charge-offs	2,460	2,505	2,901
Commercial loans			
Allowance for loan losses	1,540	1,012	169
Provision for loan losses	1,100	442	95
Charge-offs	1,488	811	717
Recoveries	428	424	673
Net charge-offs	1,060	387	44

Finally, Yoo notes the following supplementary information from Company XYZ's annual report:

- Competition in the commercial loan space has become increasingly fierce, leading XYZ managers to pursue higher-risk strategies to increase market share.
- The net benefit plan obligation has steadily decreased during the last three years.
- Company XYZ awards above-average equity-based compensation to its top managers.

8 Multiple Choice 1 / 1 point

Which of the following industries *most likely* has the highest level of global systemic risk?

A. Industry A

B. Industry B

C. Industry C

Feedback

C is correct. Industry C, representing global commercial banks, most likely has the highest level of global systemic risk because global commercial banks have the highest proportion of cross-border business. Unlike banks, the overall insurance market (of which Industry A is a subset) has a smaller proportion of cross-border business, and insurance companies' foreign branches are generally required to hold assets in a jurisdiction that are adequate to cover the related policy liabilities in that jurisdiction. As an international property and casualty (P&C) insurer, Company A provides protection against adverse events related to autos, homes, or commercial activities; many of these events have local, rather than international, impact. Industry B, credit unions, most likely has the lowest level of global systemic risk. Credit unions are depository institutions that function like banks and offer many of the same services, but they are owned by their members rather than being publicly traded as many banks are.

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9 Multiple Choice 1 / 1 point

Based on Exhibit 1, Company XYZ's capital adequacy over the last three years, as measured by the three key capital ratios, signals conditions that are:

A. mixed.

B. declining.

C. improving.

Feedback

A is correct. Company XYZ's key capital adequacy ratios show mixed conditions. The ratios are calculated as follows:

Common Equity Tier 1 Capital Ratio =

2015 Common Equity Tier 1 Capital Ratio =

2016 Common Equity Tier 1 Capital Ratio = ———————

2017 Common Equity Tier 1 Capital Ratio = ———————

Tier 1 Ratio =

——

2015 Tier 1 Ratio = ————————————

2016 Tier 1 Ratio = ————————————

2017 Tier 1 Ratio = ————————————

Total Capital Ratio = ———————————————————

2015 Total Capital Ratio = ———————

2016 Total Capital Ratio = ———————

2017 Total Capital Ratio = ———————

	2017	**2016**	**2015**
Common equity Tier 1 capital ratio	11.3%	11.1%	11.0%
Tier 1 capital ratio	13.0%	12.7%	12.5%
Total capital ratio	14.7%	14.8%	15.5%

The common equity Tier 1 capital ratio and the Tier 1 capital ratio both strengthened from 2015 to 2017, but the total capital ratio weakened during that same period, signaling mixed conditions.

10 Multiple Choice 1 / 1 point

Based only on Exhibit 2, asset composition from 2015 to 2017 indicates:

- A. declining liquidity.
- B. increasing risk based on the proportion of total loans to total assets.
- C. decreasing risk based on the proportion of investments to total assets.

Feedback

A is correct. Company XYZ's liquid assets as a percentage of total assets declined each year since 2015, indicating declining liquidity.

	2017		2016		2
	$m	% of Total Assets	$m	% of Total Assets	$m
Total liquid assets	361,164	18.7%	354,056	19.7%	356,25
Investments	434,256	22.5%	367,158	20.4%	332,46
Loans					
Consumer loans	456,957		450,576		447,49
Commercial loans	499,647		452,983		403,05
Total loans	956,604	49.6%	903,559	50.3%	850,55
Goodwill	26,693	1.4%	26,529	1.5%	25,705
Other assets	151,737	7.9%	144,210	8.0%	121,78
Total assets	1,930,454	100%	1,795,512	100%	1,686,75

11 Multiple Choice 1 / 1 point

Based on Exhibit 3, the trend in impairment allowances is reflective of the changes in:

A. impaired assets.

B. strong credit quality assets.

C. past due but not impaired assets.

Feedback

C is correct. Impairment allowances have increased proportionately to the increases in the amount of past due but not impaired assets, which may be in anticipation of these past due assets becoming impaired. Impaired assets have decreased each year while strong credit quality assets have increased each year, which suggests lowering impairment allowances as a result of improving credit quality of these financial instruments.

At 31 December	2017	2016	2015
	$m	$m	$m
Strong credit quality	338,948	327,345	320,340
Good credit quality	52,649	54,515	54,050
Satisfactory credit quality	51,124	55,311	56,409
Substandard credit quality	23,696	24,893	27,525
Past due but not impaired	2,823	2,314	2,058
Impaired	8,804	9,345	10,235
Total gross amount	478,044	473,723	470,617
Impairment allowances	−5,500	−4,500	−4,000
Total	472,544	469,223	466,617
YoY change in impaired assets	−5.8%	−8.7%	
YoY change in strong credit quality assets	3.5%	2.2%	
YoY change in past due but not impaired assets	22.0%	12.4%	
YoY change in impairment allowances	22.2%	12.5%	

Note: YoY = year-over-year

2015 to 2016 change in impaired assets: ______

2015 to 2016 change in strong credit quality assets: ______

2015 to 2016 change in past due but not impaired assets: ______

2015 to 2016 change in impairment allowances: ______

2016 to 2017 change in impaired assets: ______

2016 to 2017 change in strong credit quality assets: ______

2016 to 2017 change in past due but not impaired assets: ______

2016 to 2017 change in impairment allowances: ______

12 Multiple Choice 0 / 1 point

Based on Exhibit 4, a loan loss analysis for the last three years indicates that:

- ✕ A. Company XYZ has become less conservative in its provisioning for consumer loans.

Correct Answer: **C. the cushion between the allowance and the net commercial loan charge-offs has declined.**

- B. the provision for commercial loan losses has trailed the actual net charge-off experience.
- C. the cushion between the allowance and the net commercial loan charge-offs has declined.

Feedback

C is correct. The allowance for loan losses to net commercial loan charge-offs has been declining during the last three years, which indicates that the cushion between the allowance and the net commercial loan charge-offs has deteriorated.

2015 Consumer:

2016 Consumer:

2017 Consumer:

2015 Commercial:

2016 Commercial:

2017 Commercial:

2015 Consumer:

2016 Consumer:

2017 Consumer:

2015 Commercial:

2016 Commercial:

2017 Commercial:

	2017	2016	2015
	$m	$m	$m
Consumer loans			
Allowance for loan losses	11,000	11,500	13,000
Provision for loan losses	3,000	2,000	1,300
Charge-offs	3,759	3,643	4,007
Recoveries	1,299	1,138	1,106
Net charge-offs	2,460	2,505	2,901
Commercial loans			
Allowance for loan losses	1,540	1,012	169
Provision for loan losses	1,100	442	95
Charge-offs	1,488	811	717
Recoveries	428	424	673
Net charge-offs	1,060	387	44
Allowance for loan losses to net loan charge-offs: consumer	4.47	4.59	4.48
Allowance for loan losses to net loan charge-offs: commercial	1.45	2.61	3.84
Provision for loan losses to net loan charge-offs: consumer	1.22	0.80	0.45
Provision for loan losses to net loan charge-offs: commercial	1.04	1.14	2.16

13 Multiple Choice 0 / 1 point

Which of the following supplemental factors is consistent with a favorable assessment of Company XYZ’s financial outlook?

× A. Competitive environment

Correct Answer: **B. Net benefit plan obligation**

B. Net benefit plan obligation

C. Equity-based compensation policy

Feedback

B is correct. The net benefit plan obligation has steadily decreased during the last three years, which indicates a lower degree of risk posed by the benefit plan.

Vignette

Ivan Paulinic, an analyst at a large wealth management firm, meets with his supervisor to discuss adding financial institution equity securities to client portfolios. Paulinic focuses on Vermillion Insurance (Vermillion), a property and casualty company, and Cobalt Life Insurance (Cobalt). To evaluate Vermillion further, Paulinic compiles the information presented in Exhibit 1.

Exhibit 1
Select Financial Ratios for Vermillion Insurance

Ratio	**2017**	**2016**
Loss and loss adjustment expense	59.1%	61.3%
Underwriting expense	36.3%	35.8%
Combined	95.4%	97.1%
Dividend	2.8%	2.6%

In addition to the insurance companies, Paulinic gathers data on three national banks that meet initial selection criteria but require further review. This information is shown in Exhibits 2, 3, and 4.

Exhibit 2

Select Balance Sheet Data for National Banks—Trading: Contribution to Total Revenues

Bank	2017	2013	2009	2005
N-bank	4.2%	7.0%	10.1%	8.9%
R-bank	8.3%	9.1%	17.0%	7.9%
T-bank	5.0%	5.0%	11.9%	6.8%

Focusing on N-bank and T-bank, Paulinic prepares the following data.

Exhibit 3
2017 Select Data for N-bank and T-bank

	N-bank		T-bank	
	2017	2016	2017	2016
Average daily trading VaR ($ millions)	11.3	12.6	21.4	20.5
Annual trading revenue/average daily trading VaR	160×	134×	80×	80×

Paulinic investigates R-bank's risk management practices with respect to the use of credit derivatives to enhance earnings, following the 2008 financial crisis. Exhibit 4 displays R-bank's exposure over the last decade to credit derivatives not classified as hedges.

Exhibit 4
R-bank's Exposure to Freestanding Credit Derivatives

Credit Derivative Balances	2017	2012	2007
Notional amount ($ billions)	13.4	15.5	305.1

All of the national banks under consideration primarily make long-term loans and source a significant portion of their funding from retail deposits. Paulinic and the rest of the research team note that the central bank is unwinding a long period of monetary easing as evidenced by two recent increases in the overnight funding rate. Paulinic informs his supervisor that:

Statement 1:

Given the recently reported stronger-than-anticipated macroeconomic data, there is an imminent risk that the yield curve will invert.

Statement 2:

N-bank is very active in the 30-day reverse repurchase agreement market during times when the bank experiences significant increases in retail deposits.

14 Multiple Choice 1 / 1 point

Paulinic's analysis of the two insurance companies *most likely* indicates that:

- A. Cobalt has more-predictable claims than Vermillion.
- B. Cobalt has a higher capital requirement than Vermillion.
- C. Vermillion's calculated risk-based capital is more sensitive than Cobalt's to interest rate risk.

Feedback

A is correct. Claims associated with life and health insurance companies (Cobalt) are more predicable than those for property and casualty insurance companies (Vermillion). Property and casualty insurers' claims are more variable and "lumpier" because they arise from accidents and other unpredictable events, whereas life and health insurers' claims are more predictable because they correlate closely with relatively stable actuarially based mortality rates when applied to large populations.

15 Multiple Choice 1 / 1 point

Based only on the information in Exhibit 1, in 2017 Vermillion *most likely*:

- A. experienced a decrease in overall efficiency.

B. improved its ability to estimate insured risks.

C. was more efficient in obtaining new premiums.

Feedback

B is correct. The loss and loss adjustment expense ratio decreased from 61.3% to 59.1% between 2016 and 2017. This ratio is calculated as follows: (Loss Expense + Loss Adjustment Expense)/Net Premiums Earned. The loss and loss adjustment expense ratio indicates the degree of success an underwriter has achieved in estimating the risks insured. A lower ratio indicates greater success in estimating insured risks.

16 Multiple Choice 0 / 1 point

Based only on Exhibit 2, which of the following statements is correct?

A. The quality of earnings for R-bank was the highest in 2009.

B. Relative to the other banks, N-bank has the highest quality of earnings in 2017.

C. Trading represented a sustainable revenue source for T-bank between 2005 and 2013.

Correct Answer: **B. Relative to the other banks, N-bank has the highest quality of earnings in 2017.**

Feedback

B is correct. The quality of earnings is directly related to the level of sustainable sources of income. Trading income tends to be volatile and not necessarily sustainable. Higher-quality income would be net interest income and fee-based service income. Because N-bank's 2017 trading revenue contribution is the lowest relative to other banks, its quality of earnings would be considered the best of the three banks.

17 Multiple Choice 1 / 1 point

Based only on Exhibit 3, Paulinic should conclude that:

A. trading activities are riskier at T-bank than N-bank.

B. trading revenue per unit of risk has improved more at N-bank than T-bank.

C. compared with duration, the metric used is a better measure of interest rate risk.

Feedback

B is correct. Trading revenue per unit of risk can be represented by the ratio of annual trading revenue to average daily trading value at risk (VaR) and represents a measure of reward-to-risk. The trading revenue per unit of risk improved at N-bank (from 134× to 160×) between 2016 and 2017, and there was no change at T-bank (80×). VaR can be used for gauging trends in intra-company risk taking.

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18 Multiple Choice 1 / 1 point

Based only on Exhibit 4, R-bank's use of credit derivatives since 2007 *most likely*:

A. increased posted collateral.

✓ B. decreased the volatility of earnings from trading activities.

C. indicates consistent correlations among the relevant risks taken.

Feedback

B is correct. Exhibit 4 indicates that exposure to free-standing credit derivatives dramatically declined from a peak during the global financial crisis in 2008. If a derivatives contract is classified as freestanding, changes in its fair value are reported as income or expense in the income statement at each reporting period. The immediate recognition of a gain or loss in earnings, instead of reporting it in other comprehensive income, can lead to unexpected volatility of earnings and missed earnings targets. As a result, earnings volatility from the use of credit derivatives most likely decreased.

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19 Multiple Choice 0 / 1 point

Based on Statement 1, the net interest margin for the three banks' *most likely* will:

A. decrease.

B. remain unchanged.

✗ C. increase.

Correct Answer: **A. decrease.**

Feedback

A is correct. A bank's net interest revenue represents the difference between interest earned on loans and other interest-bearing assets and the level of interest paid on deposits and other interest-bearing liabilities. Banks typically borrow money for shorter terms (retail deposits) and lend to customers for longer periods (mortgages and car loans). If the yield curve unexpectedly inverts, the short-term funding costs will increase and the net interest margin will most likely decrease (not remain unchanged or increase).

20 Multiple Choice 1 / 1 point

Based on Statement 2, the financial ratio *most* directly affected is the:

A. Tier 2 capital ratio.

B. net stable funding ratio.

C. liquidity coverage ratio.

Feedback

C is correct. Reverse repurchase agreements represent collateralized loans between a bank and a borrower. A reverse repo with a 30-day maturity is a highly liquid asset and thus would directly affect the liquidity coverage ratio (LCR). LCR evaluates short-term liquidity and represents the percentage of a bank's expected cash outflows in relation to highly liquid assets.

Vignette

Mindy Chen manages a global equity portfolio for First Dawn Investment Management. Chen analyzes Great White North Bank (GWNB) for possible inclusion in the portfolio. Chen uses the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach with an emphasis on capital adequacy and asset quality.

Chen begins her analysis by assessing GWNB's capital adequacy using the data in Exhibit 1.

Exhibit 1 (in USD millions)

Regulatory Capital and Risk-Weighted Assets	Current Year	Prior Year
Common Equity Tier 1 capital	17,252	14,783
Additional Tier 1 Capital	1,879	1,927
Tier 2 Capital	1,554	988
Risk-weighted assets	218,617	212,587

Chen recalls that the minimum total capital ratio under Basel III is 8%. Chen then analyzes GWNB's asset quality using the data in Exhibit 2.

Exhibit 2 (in USD millions)

Assets	Current Year	Prior Year
Liquid assets	62,809	41,474
Investments	14,602	14,481
Loans	211,537	220,828
Allowance for loan losses	17,500	18,600
Tangible assets	32,180	32,405
Intangible assets	13,798	13,482
Deferred tax assets	4,897	4,735
Total assets	357,323	346,005

Chen considers the proportionate increases in the following income statement items and how they would impact GWNB's CAMELS rating:

- Item 1: Trading income
- Item 2: Service income
- Item 3: Net interest income

21 Multiple Choice 1 / 1 point

Based on Exhibit 1, compared to the prior year, GWNB's current year Total Tier 1 Capital ratio *most likely* has a net increase *closest* to:

0.89%.

Correct Answer Feedback: Correct because GWNB's Total Tier 1 Capital ratio in the prior year = (14,783 + 1,927) / 212,587 ≈ 0.0786 = 7.86%. In the current year GWNB's Total Tier 1 Capital ratio = (17,252 + 1,879) / 218,617 ≈ 0.0875 = 8.75%. GWNB's Total Tier 1 Capital ratio therefore increased by 8.75% – 7.86% = 0.89% in the current year.

0.94%.

Incorrect Answer Feedback: Incorrect because this is the increase in GWNB's Common Equity Tier 1 Capital ratio. In the prior year = 14,783 / 212,587 ≈ 0.0695 = 6.95%. In the current year GWNB's Total Tier 1 Capital ratio = 17,252 / 218,617 ≈ 0.0789 = 7.89%. GWNB's Common Equity Tier 1 Capital ratio therefore increased by 7.89% – 6.95% = 0.94% in the current year.

1.13%.

Incorrect Answer Feedback: Incorrect because this is the increase in GWNB's Total Capital ratio. Total Capital ratio in the prior year = (14,783 + 1,927 + 988) / 212,587 ≈ 0.0833 = 8.33%. In the current year GWNB's Total Capital ratio = (17,252 + 1,879 + 1,554) / 218,617 ≈ 0.0946 = 9.46%. GWNB's Total Capital ratio therefore improved in the current year by 9.46% – 8.33% = 1.13%.

22 Multiple Choice 1 / 1 point

Did GWNB meet the minimum total capital requirement under Basel III in both the current and prior years?

Yes.

Correct Answer Feedback: Correct because GWNB's minimum total capital requirement exceeded 8% in both the prior and current years. To calculate the ratio in each year we use: (common equity tier 1 capital + additional tier 1 capital + tier 2 capital) / (risk-weighted assets). For the prior year GWNBs ratio = (14,783 + 1,927 + 988) / 212,857 ≈ 8.3% which exceeds the 8% minimum. For the current year GWNBs ratio = (17,252 + 1,879 + 1,554) / 218,617 ≈ 9.5% which exceeds the 8% minimum.

No, GWNB was below the minimum Basel III total capital requirement in the prior year.

Incorrect Answer Feedback: Incorrect because in the prior year GWNBs ratio = (14,783 + 1,927 + 988) / 212,857 ≈ 8.3% which exceeds the 8% minimum.

No, GWNB was below the minimum Basel III total capital requirement in the current year.

Incorrect Answer Feedback: Incorrect because in the current year GWNBs ratio = (17,252 + 1,879 + 1,554) / 218,617 ≈ 9.5% which exceeds the 8% minimum.

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23 Multiple Choice 1 / 1 point

Based on Exhibit 2, compared to the prior year GWNB's asset quality in the current year *most likely*:

decreased.

Incorrect Answer Feedback: Incorrect because GWNB's liquid assets grew relative to total assets from 41,474 / 346,005 ≈ 12% in the prior year to 62,809 / 357,323 ≈17.6% in the current year, while loans decreased relative to total assets from 220,828 / 346,005 ≈ 63.8% to 211,537 / 357,323 ≈ 59.2%. Other asset categories were relatively unchanged year over year. Both the reduction in loans and the increase in liquid assets as a percent of total assets suggest an improvement in asset quality.

stayed the same.

Incorrect Answer Feedback: Incorrect because GWNB's liquid assets grew relative to total assets from 41,474 / 346,005 ≈ 12% in the prior year to 62,809 / 357,323 ≈17.6% in the current year, while loans decreased relative to total assets from 220,828 / 346,005 ≈ 63.8% to 211,537 / 357,323 ≈ 59.2%. Other asset categories were relatively unchanged year over year. Both the reduction in loans and the increase in liquid assets as a percent of total assets suggest an improvement in asset quality.

increased.

Correct Answer Feedback: Correct because GWNB's liquid assets grew relative to total assets from 41,474 / 346,005 ≈ 12% in the prior year to 62,809 / 357,323 ≈17.6% in the current year, while loans decreased relative to total assets from 220,828 / 346,005 ≈ 63.8% to 211,537 / 357,323 ≈ 59.2%. Other asset categories were relatively unchanged year over year. Both the reduction in loans and the increase in liquid assets as a percent of total assets suggest an improvement in asset quality.

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24 Multiple Choice 1 / 1 point

Which of the income statement items that Chen considers would *most likely* result in an improved CAMELs rating?

Item 1 and Item 2

Incorrect Answer Feedback: Incorrect because trading income (Item 1) for banks tends to be more volatile and therefore of lower quality.

- Item 1 and Item 3

Incorrect Answer Feedback: Incorrect because trading income (Item 1) for banks tends to be more volatile and therefore of lower quality.

- Item 2 and Item 3

Correct Answer Feedback: Correct because service income (Item 2) and net interest income (Item 3) provide more sustainable earnings than does trading income (Item 1). Regarding sustainability of a bank's earnings, it is important to examine the composition of earnings. Banks' earnings typically comprise (a) net interest income (the difference between interest earned on loans minus interest paid on the deposits supporting those loans), (b) service income, and (c) trading income. Of these three general sources, trading income is typically the most volatile. Thus, a greater proportion of net interest income and service income is typically more sustainable than trading income.

Vignette

Matthew Green, a research analyst, is asked to evaluate the performance of Second National Bank (SNB), a US commercial bank. Green prepares a research report to support his recommendation.

Green notices that SNB's CEO was appointed 18 months ago. Previously, the CEO was the president of a non-profit, non-public depository institution that offered a comparable selection of deposits, loans, and mortgages as SNB.

Green includes the following Basel III highlights in his research report:

- Highlight 1 : Basel III specifies the minimum percentage of its risk-weighted assets that a bank must fund with equity capital.
- Highlight 2 : Basel III specifies that a bank must hold enough high-quality liquid assets to cover its liquidity needs in a 30-day liquidity stress scenario.
- Highlight 3 : Basel III requires a bank to have a minimum amount of stable funding relative to the bank's liquidity needs in a 90-day liquidity stress scenario.

To assess the ability of SNB to meet Basel III minimum capital requirements, Green calculates SNB's Total Tier 1 Capital. Green reviews the latest financial disclosures, and includes retained earnings, accumulated other comprehensive income, and all instruments with a fixed maturity that are subordinate to depositors in his calculation.

Green notices that the allowance for loan losses decreased year over year as a result of a recovery of a prior fiscal year charge-off. Green calculates several comparative ratios to assess the quality of the allowance for loan losses.

25 Multiple Choice 1 / 1 point

Which of the following *best* describes the CEO's previous financial institution?

- Mutual bank

Correct Answer Feedback: Correct because credit unions, cooperative and mutual banks are depository institutions that function like banks and offer many of the same services as banks. They are owned by their members, rather than being publicly traded like many banks. Another difference from commercial banks is that these institutions are organized as non-profits and, therefore, do not pay income taxes.

- Mortgage bank

  Incorrect Answer Feedback: Incorrect because mortgage banks originate, sell, and service mortgages and are usually active participants in the securitization markets.

- Savings and loan association

  Incorrect Answer Feedback: Incorrect because savings and loan associations are depository institutions that specialize in financing long-term residential mortgages.

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26 Multiple Choice 1 / 1 point

Which of the following highlights included in Green's research report is incorrect?

- Highlight 1

  Incorrect Answer Feedback: Incorrect because Basel III specifies the minimum percentage of its risk-weighted assets that a bank must fund with equity capital. Therefore, Highlight 1 is an accurate statement.

- Highlight 2

  Incorrect Answer Feedback: Incorrect because Basel III specifies that a bank must hold enough high-quality liquid assets to covers its liquidity needs in a 30-day liquidity stress scenario. Therefore, Highlight 2 is an accurate statement.

- Highlight 3 ✓

  Correct Answer Feedback: Correct because Basel III requires a bank to have a minimum amount of stable funding relative to the bank's liquidity needs over a one-year horizon and not a 90-day liquidity stress scenario. Therefore, Highlight 3 is an inaccurate statement.

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27 Multiple Choice 1 / 1 point

In calculating SNB's Total Tier 1 Capital, Green is incorrect in regard to:

- retained earnings.

  Incorrect Answer Feedback: Incorrect because common Equity Tier 1 Capital includes common stock, issuance surplus related to common stock, retained earnings, accumulated other comprehensive income, and certain adjustments including the deduction of intangible assets and deferred tax assets. Other Tier 1 Capital includes other types of instruments issued by the bank that meet certain criteria. The criteria require, for example, that the instruments be subordinate to such obligations as deposits and other debt obligations, not have a fixed maturity, and not have any type of payment of dividend or interest that is not totally at the discretion of the bank. Therefore, this response should be included in SNB's Total Tier 1 Capital.

- accumulated other comprehensive income.

Incorrect Answer Feedback: Incorrect because common Equity Tier 1 Capital includes common stock, issuance surplus related to common stock, retained earnings, accumulated other comprehensive income, and certain adjustments including the deduction of intangible assets and deferred tax assets. Other Tier 1 Capital includes other types of instruments issued by the bank that meet certain criteria. The criteria require, for example, that the instruments be subordinate to such obligations as deposits and other debt obligations, not have a fixed maturity, and not have any type of payment of dividend or interest that is not totally at the discretion of the bank. Therefore, this response should be included in SNB's Total Tier 1 Capital.

instruments that are subordinate to depositors.

Correct Answer Feedback: Correct because instruments that are subordinate to depositors and to general creditors of the bank must not have a fixed maturity to be included in Other Tier 1 Capital. Other Tier 1 Capital includes other types of instruments issued by the bank that meet certain criteria. The criteria require, for example, that the instruments be subordinate to such obligations as deposits and other debt obligations, not have a fixed maturity, and not have any type of payment of dividend or interest that is not totally at the discretion of the bank. Therefore, this response should not be included in SNB's Total Tier 1 Capital.

28 Multiple Choice 1 / 1 point

The comparative ratio that Green is *least likely* to calculate is the ratio of the allowance for loan losses to:

net income.

Correct Answer Feedback: Correct because this ratio would not provide any insight to the quality of the allowance for loan losses as it would only represent a comparison against the overall net income of the financial institution.

net loan charge-offs.

Incorrect Answer Feedback: Incorrect because it is one of three ratios are helpful in assessing the quality of the allowance for loan losses:

- the ratio of the allowance for loan losses to non-performing loans

- the ratio of the allowance for loan losses to net loan charge-offs

- the ratio of the provision for loan losses to net loan charge-offs

non-performing loans.

Incorrect Answer Feedback: Incorrect because it is one of three ratios are helpful in assessing the quality of the allowance for loan losses:

- the ratio of the allowance for loan losses to non-performing loans

- the ratio of the allowance for loan losses to net loan charge-offs

- the ratio of the provision for loan losses to net loan charge-offs

Vignette

Aiko Takahashi works as a senior examiner for the Financial Institutions Agency (FIA), and Haruto Suzuki is one of the analysts who reports to her. Suzuki is a new hire who previously worked at an investment firm that specialized in manufacturing companies. His first two weeks at the FIA were spent in training sessions focused on the Basel III regulatory framework for banks to prepare him for his new analyst role.

At today's weekly meeting, Takahashi and her analysts are conducting a preliminary review of some of their assigned insurance institutions. Suzuki arrives early, so Takahashi takes the opportunity to ask him to explain his understanding of the Basel III framework. Suzuki describes three enhancements that Basel III incorporated as part of its transition from the previous iteration:

- Enhancement 1: Additional measures to ensure sufficient earnings quality, thereby strengthening banks' regulatory capital.
- Enhancement 2: More stringent capital requirements that prevent banks from taking on excessive financial leverage.
- Enhancement 3: Increased funding stability by giving preference to interbank deposits over those from consumers.

The other analysts arrive, and the meeting gets underway. The group is scheduled to review two insurance companies: Yatabe Holdings (YH) and Sapporo Hoken (SH). Takahashi focuses first on YH, a property and casualty insurer. Takahashi notes that the property and casualty market has been in a "hard" phase for a few years now, and she wonders whether the results will show the beginning of a cyclical shift.

The group reviews YH's profitability information that Suzuki has provided in Exhibit 1. Takahashi asks Suzuki to calculate YH's combined ratio:

Exhibit 1 Excerpts from YH's Financial Statements

(¥ billions)	
Loss expense and loss adjustment expense	38,467
Net premiums earned	62,939
Net premiums written	49,241
Underwriting expense	23,960

Glancing at her watch, Takahashi notes that they don't have enough time to look at SH. She comments that SH has a lot of international business and suggests they finish this discussion at next week's meeting.

29 Multiple Choice 0 / 1 point

Which of the enhancements Suzuki describes regarding the Basel III regulatory framework is *most* accurate?

- Enhancement 1

  Incorrect Answer Feedback: Incorrect. Earnings quality was not a focus of Basel III, though the changes implemented under Basel III will likely lead to improved earnings quality.

- Enhancement 2

  Correct Answer Feedback: Correct. In the aftermath of the 2008 crisis, one of the key changes in Basel III was an increase in the required level of equity funding of banks' risk-weighted assets. This measure essentially prevents banks from assuming excessive financial leverage.

Enhancement 3

Incorrect Answer Feedback: Incorrect. Although Basel III did increase requirements for stable funding, deposits from consumers are considered more stable than those raised in the interbank markets. Therefore, increased funding stability is achieved by giving preference to consumer deposits, not interbank deposits.

30 Multiple Choice 1 / 1 point

If Takahashi's interpretation of the current phase of the property and casualty market is correct, a cyclical shift is *most likely* to result in:

a tightening of underwriting standards.

Incorrect Answer Feedback: Incorrect. In hard insurance markets, premiums and profitability are high and insurers can set high underwriting standards. As competitors are attracted to this profitability and the market softens, underwriting standards and premiums drop as everyone offers the cheapest deals.

capital adequacy being easier to achieve.

Incorrect Answer Feedback: Incorrect. Hard markets feature higher premium prices, increased profitability, and an increased ability for insurers to maintain adequate capital. Capital typically depletes as markets soften, making capital adequacy more difficult to achieve.

additional competitors entering the market.

Correct Answer Feedback: **Correct.** In hard insurance markets, insurance premiums and industry profits are high as are underwriting standards. This attracts competitors that offer lower premiums resulting in lower profitability, a softening of the market, and a weakening of underwriting standards.

31 Multiple Choice 0 / 1 point

Based on Exhibit 1, YH's combined ratio is *closest* to:

61.1%.

Incorrect Answer Feedback: Incorrect because this is the loss Expenses & loss adjustment expense ratio (in ¥ billions, except %):

Loss Expense & Loss Adjustment Expense Ratio	
Loss expense & loss adjustment expense	38,467
Net premiums earned	62,939
Ratio	61.1%

109.8%.

Correct Answer Feedback: **Correct.** Loss Expense & Loss Adjustment expense ratio and the underwriting expense ratio (in ¥ billions, except %):

Loss Expense & Loss Adjustment Expense Ratio (A)		Underwriting Expense Ratio (B)		Combined Ratio (A+B)
Loss expense & loss adjustment expense	38,467	Underwriting expense	23,960	
Net premiums earned	62,939	Net premiums written	49,241	
Ratio (A)	61.1%	Ratio (B)	48.7%	109.8%

126.8%.

Incorrect Answer Feedback: Incorrect. In this calculation, the loss expense & loss adjustment expense ratio incorrectly uses net premiums written as the denominator (in ¥ billions, except %):

Incorrect **Loss Expense & Loss Adjustment Expense Ratio (A)**		Underwriting Expense Ratio (B)		Combined Ratio (A+B)
Loss expense & loss adjustment expense	38,467	Underwriting expense	23,960	
Net premiums written *(incorrect)*	49,241	Net premiums written	49,241	
Ratio (A)	78.1%	Ratio (B)	48.7%	126.8%

Multiple Choice 1 / 1 point

Based solely on the comment Takahashi makes in postponing the review of SH to the next meeting, the primary line of insurance that SH offers is *most likely*:

reinsurance.

Correct Answer Feedback: **Correct.** Reinsurance business is largely international, while the rest of the insurance market has a smaller proportion of cross-border business.

life and health.

Incorrect Answer Feedback: Incorrect. Life and health insurance companies tend to have a smaller proportion of cross-border business. Reinsurance companies are much more likely to have significant international business.

property and casualty.

Incorrect Answer Feedback: Incorrect. Property and casualty insurance companies tend to have a smaller proportion of cross-border business. Reinsurance companies are much more likely to have significant international business.

Vignette

Suri Reddy, a junior equity analyst at Sutton Management Inc. (Sutton), is currently on rotation in the Financial Institutions sub-group at Sutton to assist in the evaluation of First West Bank (First West), a small regional bank in the northwestern United States.

In preparation for her first meeting with her new team, Reddy makes the following notes about how she thinks banks differ from the industrial companies on which she usually works.

- Banks will have fewer tangible assets.
- Bank assets are predominantly measured at fair market value.
- Because banks do not carry inventory, liquidity risks will be negligible.

Sutton uses the CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) approach to analyzing banks, and Reddy is assigned to look at First West's capital adequacy. Exhibit 1 contains the capital adequacy ratios for First West for the past five years and the Basel III guidelines.

Exhibit 1 Capital Adequacy Ratios for First West Bank and Basel III Requirements

	Basel III	Current Year	Previous Year	2 Years Prior	3 Years Prior	4 Years Prior
Common equity tier 1 ratio	4.5%	9.5%	9.1%	8.5%	8.0%	8.0%
Total tier 1 capital ratio	6.0	10.7	11.0	9.7	9.3	9.7
Total capital ratio	8.0	12.5	13.1	12.7	12.8	13.9

To further understand the changes in the Current Year, Reddy retrieves the detailed components of the ratios for the past two years, shown in Exhibit 2.

Exhibit 2 Details of Capital Adequacy Ratios for First West Bank for the Current Year and the Previous Year (in $ thousands)

Common equity tier 1	**Current Year**	**Previous Year**
Qualifying common shares	756,800	749,600
Retained earnings	1,488,600	1,355,000
Accumulated other comprehensive income	(29,000)	(32,700)
	2,216,400	2,071,900
Regulatory adjustments	(206,800)	(208,600)
Common equity tier 1	2,009,600	1,863,300
Tier 1 capital	2,274,800	2,233,400
Total capital (tier 1 & 2)	2,644,800	2,669,400
Risk-weighted assets	21,180,100	20,364,500

Finally, Reddy reviews First West's annual report for background information on non-CAMELS factors that will be relevant to the analysis. First West focuses on loans to small and mid-sized regional businesses. Its mission is to provide a comprehensive approach to business owners of the region by primarily serving their domestic needs for business, personal banking, and wealth management. Using a relationship-based approach, First West holds the dominant position in its regional market. Reddy considers how these factors affect risk exposure for First West relative to global competitors that are considering entering its market.

33 Multiple Choice 1 / 1 point

Which of Reddy's notes concerning banks versus industrials is *least* accurate? The note concerning:

- fair value.

  Incorrect Answer Feedback: Incorrect. Reddy's note about fair value is correct. Banks' assets are primarily financial assets (loans) and are often measured at fair value.

- liquidity risks.

  Correct Answer Feedback: **Correct.** Liquidity risk is a major concern for a bank, despite not carrying inventory. A bank's assets are primarily financial assets (loans), which are often measured at fair value. It is important that a bank have adequate liquidity in its assets such that it could withstand the demand for a redemption of deposits (liabilities) if it arose.

- tangible assets.

  Incorrect Answer Feedback: Incorrect. Reddy's note about tangible assets is correct. Banks generally do have fewer tangible assets than industrials.

34 Multiple Choice 1 / 1 point

Based on the most important metric in Exhibit 1, the *best* conclusion about First West's capital adequacy over the past five years is that it has:

- declined.

  Incorrect Answer Feedback: Incorrect. The total capital ratio decreased over the period, but it is not considered the most important metric in assessing capital adequacy.

- improved.

  Correct Answer Feedback: **Correct.** The most important metric in assessing capital adequacy is the common equity tier 1 ratio. First West improved its position in that ratio by 150 bps in the five-year period (8.0% to 9.5%).

- been unstable.

  Incorrect Answer Feedback: Incorrect. The total capital ratio has been unstable (decreasing, increasing, and then decreasing again), but it is not considered the most important metric in assessing capital adequacy.

35 Multiple Choice 1 / 1 point

Using Exhibits 1 and 2, the greatest effect on the improvement in the common equity tier 1 ratio from the Previous Year to the Current Year is *best* explained by the:

- increase in retained earnings.

Correct Answer Feedback: **Correct.** The common equity tier 1 ratio is Common equity tier 1 ÷ Risk weighted assets. The increase in retained earnings, with a growth rate of 9.9%, had the greatest effect on the improvement in the common equity tier 1 ratio as shown in the table below. Although the percentage decrease in the negative amount of accumulated other comprehensive income was larger, at 11.3%, it is on a much smaller base and, therefore, would have had a smaller effect. The risk-weighted assets increased, which would decrease the ratio because it is the denominator.

Change in Components of the Common Equity Tier 1 Ratio (in $ thousands)

	Current Year	Previous Year	Change	Growth
Qualifying common shares	756,800	749,600	7,200	1.0%
Retained earnings	1,488,600	1,355,000	133,600	**9.9%**
Accumulated other comprehensive income	(29,000)	(32,700)	3,700	(11.3)%
	2,216,400	2,071,900	144,500	7.0%
Regulatory adjustments	(206,800)	(208,600)		
Common equity tier 1	2,009,600	1,863,300	146,300	7.9%
Risk-weighted assets	21,180,100	20,364,500	815,600	4.0%

increase in risk-weighted assets.

Incorrect Answer Feedback: Incorrect. The risk-weighted assets had the largest dollar increase, but it would decrease the ratio because it is the denominator.

decrease in accumulated other comprehensive income.

Incorrect Answer Feedback: Incorrect. Although the decrease in the negative amount of accumulated other comprehensive income was larger, at 11.3%, than the increase in retained earnings, it is on a much smaller base and, therefore, would have had a smaller effect.

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36 Multiple Choice 1 / 1 point

Which of the following non-CAMELS factors would be *least* relevant to Reddy's analysis of First West?

Currency exposure

Correct Answer Feedback: **Correct.** First West is a regional bank focusing on the northwestern United States and primarily serving the domestic needs of their clients. It is not likely exposed to large foreign currency risks associated with holding deposits or making loans in other currencies. It could be doing some FX exchange for clients but would not have large FX currency exposure. Therefore, currency exposure would be the least relevant of the indicated factors related to her analysis. The mission of First West is to serve the business owners of the northwestern United States. As such, its fortunes are heavily tied to the economic factors of the area, which means it is less well diversified than larger banks. Although it currently has a dominant position in the region, the entry of global banks would increase the risk to the bank.

Mission of the bank

Incorrect Answer Feedback: Incorrect. The mission of First West is to serve the business owners of the northwestern United States. As such, its fortunes are heavily tied to the economic factors of the area, which means it is less well diversified than larger banks. Therefore, the mission of the bank is relevant to her analysis.

- Competitive position

Incorrect Answer Feedback: Incorrect. First West is the dominant bank in the region, but the entry of global banks that are considering entering the market would increase the risk to the bank. Therefore, its competitive position is relevant to her analysis.

Vignette

Joseph Cioffi and Amanda Yu, interns in the financial institutions division of an investment firm, are cleaning up outstanding issues in a few files.

They start by examining the investment allocation and returns on the investment portfolio for NANLife Group, a life and health insurance company, using data Cioffi prepared (Exhibit 1).

Exhibit 1 Investment Allocation and Returns, NANLife Group, in $ Millions

Investment portfolio as of 31 December	**2018**	**2017**
Loans and deposits	8,800	9,000
Debt securities	141,000	130,800
Equity securities	40,250	34,000
Derivative financial instruments	130	90
Total financial investments	190,180	173,890
Total investment portfolio	195,080	178,490
Investment returns for the year ending 31 December	**2018**	**2017**
Interest income	6,610	5,850
Dividend income	820	750
Rental income	175	170
Investment income	7,605	6,770
Gains (losses) on fixed-income investments	300	(125)
Gains (losses) on other investments	1,540	1,665
Total investment returns	9,445	8,310

Cioffi asks, “Why does NANLife have a much larger portion of its investment portfolio in equity securities compared with the property and casualty company we were working on yesterday?”

Yu replies, “NANLife can take greater risks with its investments for two reasons: First, its claims are relatively more predictable, and second, the claims have a shorter duration.”

Before moving on Cioffi comments,

“It was interesting to learn about the insurance industry when working on the NANLife file. Unlike the banking industry, it does not have to follow international standards for capital adequacy, but as a US-based insurer, NANLife still has some US requirements to meet.

The company also prepares its financial statements using statutory accounting rules that differ from both US GAAP and IFRS."

Next, they turn their attention to a commercial bank file that still required some work on the assessment of earnings quality, using summary earnings information prepared by Yu (Exhibit 2).

Exhibit 2 Selected Earnings Information from Bank File, in $ Thousands

	2018	2017	2016	2015	2014
Interest income	640,400	614,000	606,300	607,300	599,000
Interest expense	225,800	231,700	246,600	260,300	262,700
Net interest income	414,600	382,300	359,700	347,000	336,300
Trading income	3,200	5,140	6,600	(900)	15,430
Other income	135,400	125,200	116,500	117,200	100,300
Total income net of interest expense	553,200	512,640	482,800	463,300	452,030
Provision for credit losses	8,400	13,800	9,900	10,500	7,090
Earnings before taxes	117,600	95,800	91,600	78,800	87,211

Yu states,

"We have to finish the CAMELS (capital adequacy, asset quality, management capabilities, earnings sufficiency, liquidity position, and sensitivity to market risk) analysis on this file. In my preliminary review of the bank's earnings information, I assigned a score reflecting high quality of earnings, based on the following points:

- The trading income is volatile, but it is a small portion of the earnings.
- Net interest income has been growing at a constant rate over the period.
- The proportion of total income from net interest income has been stable in the most recent years."

Cioffi replies,

"I think we need to dig a little deeper. We need to look at the estimates that are used to determine earnings and whether the bank is using them to manage earnings before taxes (EBT). The provision for credit losses is an estimate where management can exercise wide discretion. Let's look at what portion of the 2018 increase in EBT comes from the change in that estimate."

37 Multiple Choice 0 / 1 point

The average return on fixed-income assets for NANLife in 2018 is *closest* to:

4.6%.

Incorrect Answer Feedback: Incorrect. It does not include the gains and losses on fixed-income investments:

6,610 ÷ 144,800 = 4.6%.

4.8%.

Correct Answer Feedback: **Correct.** The average return on fixed-income assets is calculated as follows: Investment income from fixed income ÷ Average fixed-income assets.

Investment income = Interest income + Gains/losses from fixed-income investments = $6,610 + $300 = $6,910.

Average fixed-income assets including debt securities + Loan and deposits = (½)($8,800 + $141,000) + (½)($9,000 + $130,800) = $144,800.

Average return = 6,910 ÷ 144,800 = 4.8%.

5.1%.

Incorrect Answer Feedback: Incorrect. It does not include the loans and deposits as fixed-income investments:

Average debt securities = ($141,000 + $130,800) × ½ = $135,900.

6,910 ÷ 135,900 = 5.1%.

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38 Multiple Choice 1 / 1 point

From 2017 to 2018, the risk related to the investment allocation for NANLife is *best* described as having shown:

a decrease.

Incorrect Answer Feedback: Incorrect. The proportion allocated to loans and deposits and debt securities has decreased, but this would decrease the overall risk of the investment portfolio because fixed-income investments are less risky than equity investments. The higher proportion allocated to equity would increase the portfolio risk.

no change.

Incorrect Answer Feedback: Incorrect. The total financial investments have not changed materially (97.4% to 97.5%), but that is not the best measure of the risk of the investment portfolio. The allocation of the portfolio between fixed-income and equity securities is a better measure of portfolio risk.

an increase.

Correct Answer Feedback: Correct. When the investment portfolio is examined using a common-size format, the proportion of the portfolio invested in equity securities has increased from 19.0% to 20.6% (see table below) whereas the proportion allocated to loans and deposits and debt securities has decreased. Equity investments are normally riskier than fixed-income investments, which would indicate that the investment portfolio's asset allocation is riskier in 2018 than in 2017.

Analysis of NANLife's Investment Portfolio Asset Allocation

	2018		**2017**	
Loans and deposits	8,800	4.5%	9,000	5.0%
Debt securities	141,000	72.3%	130,800	73.3%
Equity securities	40,250	20.6%	34,000	19.0%
Derivative financial instruments	130	0.1%	90	0.1%
Total financial investments	190,180	97.5%	173,890	97.4%
Investment properties	4,900	2.5%	4,600	2.6%
Total investment portfolio	195,080	100.0%	178,490	100.0%

39 Multiple Choice 1 / 1 point

Yu's reply to Cioffi's question concerning the higher portion of equity securities in NANLife's investment portfolio is *best* described as correct with respect to:

- both reasons.

  Incorrect Answer Feedback: Incorrect. The products of a P&C company are usually shorter duration than those of an L&H company; thus, that part of Yu's answer is incorrect.

- only the first reason. ✓

  Correct Answer Feedback: Correct. Yu is incorrect about the duration of the claims of NANLife. NANLife is a life and health insurance (L&H) company. L&H companies' claims are more predictable than those of property and casualty (P&C) companies, and their claims also have a longer duration than P&C those of companies. It is the combination of both of these factors that allows L&H companies to hold a greater proportion of equity investments than P&C companies.

- only the second reason.

  Incorrect Answer Feedback: Incorrect. The products of a P&C company are usually shorter duration than those of an L&H company; thus, that part of Yu's answer is incorrect.

40 Multiple Choice 1 / 1 point

Cioffi's comments about the regulatory requirements for NANLife are *best* described as:

- correct. ✓

  Correct Answer Feedback: Correct. Cioffi's comments about the regulatory requirements for US-based insurance companies are all correct. The insurance industry does not have a set of global regulatory standards like Basel III, but capital standards do exist in various jurisdictions, including the United States. In the United States, the NAIC (National Association of Insurance Commissioners) has established minimum capital adequacy standards. Also in the United States, insurance companies prepare financial reports according to statutory accounting rules, which differ from US GAAP and IFRS.

incorrect with respect to accounting rules.

Incorrect Answer Feedback: Incorrect. Cioffi's comments with respect to accounting rules are correct. In the United States, insurance companies prepare financial reports according to statutory accounting rules, which differ from US GAAP and IFRS.

incorrect with respect to capital adequacy requirements.

Incorrect Answer Feedback: Incorrect. Cioffi's comments with respect to capital adequacy are correct. The insurance industry does not have a set of global regulatory standards as the banking industry has with its Basel III regulations, but capital standards do exist in various jurisdictions, including the United States. In the United States, the NAIC has established minimum capital adequacy standards.

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41 Multiple Choice 1 / 1 point

Using Exhibit 2, which of Yu's points in support of the CAMELS score she assigns for earnings quality is *least* accurate? The point concerning:

trading income volatility.

Incorrect Answer Feedback: Incorrect. Her point about trading income is accurate. Although it is volatile, it is a small portion of net income in support of higher quality of earnings. Trading income is the least sustainable of the three types of income; therefore, low reliance on it is better for earnings quality.

growth in net interest income.

Correct Answer Feedback: Correct. Yu's point about the constant rate of growth of net interest income is not accurate. As shown in the table below (Part A), the growth rate for net interest income has been increasing, not stable, over the five-year period. Her point about trading income is accurate. Although it is volatile, it is a small portion of net income, which supports a rating of higher quality of earnings. Trading income is the least sustainable of the three types of income; therefore, low reliance on it is better for earnings quality. As shown in Part B of the table below, the mix between net interest income and other income has been stable over the last three years, which is also a sign in support of high-quality earnings.

Analysis of Bank Income

	2018	2017	2016	2015	2014
Part A					
Year-to-year growth rate					
Net interest income	8.4%	6.3%	3.7%	3.2%	---
Part B					
Portion of total income net of interest expense					
Trading income	0.6%	1.0%	1.4%	–0.2%	3.4%
Net interest income	74.9%	74.6%	74.5%	74.9%	74.4%
Other income	24.5%	24.4%	24.1%	25.3%	22.2%
Total income net of interest expense	100.0%	100.0%	100.0%	100.0%	100.0%

net interest income proportion.

Incorrect Answer Feedback: Incorrect. As shown in Part B of the table above, the mix between net interest income and other income has been stable over the last three years, which is also a sign in support of high-quality earnings.

Multiple Choice 1 / 1 point

Based on Exhibit 2, the proportion of the increase in the 2018 earnings before taxes that comes from the change in the provision for credit losses is *closest* to:

- 4.6%.

  Incorrect Answer Feedback: Incorrect; it is the change in the provision ($5,400) as a proportion of EBT but does not consider the change in EBT: 5,400 ÷ 117,600 = 4.6%.

- 7.1%.

  Incorrect Answer Feedback: Incorrect; it is the provision for credit losses as a proportion of EBT but does not consider the change in the two values: $8,400 ÷ $117,600 = 7.1%.

- 24.8%.

  Correct Answer Feedback: **Correct.** A decrease in the provision for credit losses would have the effect of increasing earnings before taxes. As shown in the table below, in 2018, EBT increased by $21,800 but the provision for credit losses decreased by $5,400, thus accounting for 24.8% of the change in EBT.

Analysis of Change in EBT and Provision for Credit Losses, in $ Thousands

	2018	2017	Change
Provision for credit losses	8,400	13,800	(5,400)
EBT	117,600	95,800	21,800
Percentage increase in EBT from decrease in provision	---	---	24.8%

Vig[illegible]

Gi[illegible]allo works as a senior manager at the National Banking Regulatory Commission (N[illegible] She is meeting with her team today to review Banco San Bruno (BSB), a large co[illegible]cial bank with operations across the country. The central bank announced a rate hike last month, and Gallo wants to re-examine BSB's ability to withstand the changing environment. Antonio Rossi, her junior colleague, displays the summary information (Exhibit 1) he has gathered on BSB's earnings composition and comments that he thinks the earnings composition has become less volatile and more sustainable.

Exhibit 1 Key Components of BSB's Earnings

(€ millions)	Current Year	Prior Year
Net interest income	55,288	50,285
Net fee income	22,648	21,107
Net trading income	18,651	18,624
Total	96,587	90,016

Gallo further explains that BSB has been flagged by the data analytics system. The flag warns that BSB may be acting to boost its net interest income by shifting its portfolio toward more long-term assets without a corresponding shift on the liability side of the balance sheet. “I don’t like it,” said Gallo. Seeing Rossi’s puzzled look, she adds, “It’s a value creation strategy in a stable interest rate market, but it heightens their risk, especially in this environment of increasing rates.”

The team next turns to BSB’s balance sheet. Rossi displays the information he has gathered (Exhibit 2), and they use it to calculate the Total Tier 1 Capital.

Exhibit 2 Excerpts from BSB’s Year-End Balance Sheet

(€ billions)	Current Year	Prior Year
Customer deposits	502	572
Subordinated debt	23	21
Preferred shares	17	17
Common shares	12	12
Retained earnings	187	201

As the discussion draws to a close, Gallo mentions that at the next meeting, they will complete a detailed CAMELS (capital adequacy, asset quality, management, earnings, liquidity, and sensitivity) analysis of BSB.

43 Multiple Choice 1 / 1 point

Rossi’s comment about BSB’s earnings composition is *best* described as:

- correct.

  Correct Answer Feedback: Correct. Net trading income is typically considered more volatile than net interest income or net fee income. As a component of the total, BSB’s net trading income has decreased from 20.7% to 19.3% per the following table, while the sustainable components (net interest income and net fee income) have increased. Overall, BSB’s earnings composition has become less volatile and more sustainable.

(€ millions)	Current Year	(%)	Prior Year	(%)
Net interest income	55,288	57.2%	50,285	55.9%
Net fee income	22,648	23.5%	21,107	23.4%
Net trading income	18,651	19.3%	18,624	20.7%
Total	96,587	100.0%	90,016	100.0%

- incorrect with respect to volatility.

  Incorrect Answer Feedback: Incorrect. Net trading income, which is typically considered volatile income, has decreased as a proportion of the total. As such, Rossi’s comment that the earnings composition has become less volatile is correct.

- incorrect with respect to sustainability.

  Incorrect Answer Feedback: Incorrect. Both net interest income and net fee income, which are considered more sustainable sources of income, have increased as a proportion of the total. As such, Rossi’s comment that the earnings composition has become more sustainable is correct.

44 Multiple Choice 0 / 1 point

The value creation strategy Gallo refers to is *best* described as:

- maturity transformation.

  Correct Answer Feedback: Correct. When banks borrow money (liabilities) on shorter terms than the terms for lending to customers (assets), they create value by lending at higher rates than their short-term funding costs. This is called maturity transformation.

- contractual maturity mismatch.

  Incorrect Answer Feedback: Incorrect. A contractual maturity mismatch is the result of the strategy described but is not the name of the strategy itself.

- risk-weighted asset stratification.

  Incorrect Answer Feedback: Incorrect. Risk-weighted asset stratification is part of the process of assessing a bank's capital adequacy.

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45 Multiple Choice 0 / 1 point

BSB's Total Tier 1 Capital (in billions) for the current year is *closest* to:

- €199.

  Incorrect Answer Feedback: Incorrect. This calculation correctly includes the Common Equity Tier 1 components (common shares (€12) and retained earnings (€187)) for a total of €199. However, it incorrectly excludes preferred shares, which should be included as part of Total Tier 1 Capital. 12 + 187 = €199.

- €216.

  Correct Answer Feedback: Correct. Total Tier 1 Capital includes Common Equity Tier 1 Capital plus other instruments (such as preferred shares) that bear no fixed maturity and carry no requirement to pay dividends or interest without the full discretion of the bank. Common Equity Tier 1 Capital includes common stock, capital surplus, retained earnings, and other comprehensive income. BSB's Total Tier 1 Capital is as follows:

(€ billions)	
Common stock	12
Retained earnings	187
Common Equity Tier 1 Capital	199
Preferred shares	17
Total Tier 1 Capital	**€216**

- €239.

  Incorrect Answer Feedback: Incorrect. This calculation correctly includes preferred shares (€17), common shares (€12), and retained earnings (€187). However, it incorrectly also includes subordinated debt (€23), bringing the total to €239. Subordinated debt is part of Tier 2 Capital. 17 + 12 + 187 +23 = €239.

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46 Multiple Choice 1 / 1 point

In the next meeting when completing the detailed analysis suggested by Gallo, which of the following topics is *least likely* to be covered?

- Currency exposure

  Correct Answer Feedback: **Correct.** In the next meeting, they are to complete a detailed CAMELS analysis of BSB, which would not include an analysis of currency exposure. Currency exposure is relevant to the analysis of a bank, but it is not covered by the CAMELS approach.

- Governance structure

  Incorrect Answer Feedback: Incorrect. A strong governance structure is a critically important aspect of the management capabilities element of CAMELS.

- Concentration of funding

  Incorrect Answer Feedback: Incorrect. Concentration of funding is one of the liquidity-monitoring metrics described in Basel III. It is covered under the liquidity position element of CAMELS.